Prashant Bhandari

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Summary

PCB Design and Embedded Firmware Engineer with hands-on internship experience taking hardware from schematic capture to production-ready PCB across satellite, industrial, and consumer IoT domains. Currently developing onboard firmware and communication subsystems for a 1U CubeSat. Proficient in KiCad, ESP32, STM32 bare-metal C, C/C++, and MicroPython; delivered 6+ end-to-end hardware projects including a LoRa module with 7 km range, a 90%-efficient buck converter, and a 1.5U CubeSat prototype. International competition finalist and electronics lead who has coached 350+ students — combining deep technical execution with proven team leadership.

Technical Skills

PCB Design: KiCad, Proteus, Schematic Capture, PCB Layout, Buck/Boost Converter Design, DRC/ERC
Embedded Systems: ESP32, STM32 Bare-Metal C, NuttX RTOS, FreeRTOS, Arduino, ECU2000, HCM111Z, BLE, LoRa, UART/SPI/I2C
Programming: C/C++, Python, MicroPython, JavaScript
Hardware: IoT Systems, Power Electronics, QC Test Fixtures, Robotics, CubeSat Subsystems
Tools: Git, Fusion 360, 3D CAD, OLED Dashboards, Sensor Integration, OTA, JLCPCB/Gerber Workflows

Experience

Satellite Systems Intern Jun 2026 – Present
Antarikchya Pratisthan Nepal — Slippers2Sat-2 (S2S-2) *Kathmandu, Nepal*

- Develop and test embedded firmware for satellite subsystems on the S2S-2 1U CubeSat bus, a student-built mission by Antarikchya Pratisthan Nepal.
- Support the communication (COM) subsystem — command reception and telemetry downlink over the amateur UHF band.
- Assist ground station operations, link verification, and decoding of downlinked mission data; debug and validate firmware during subsystem bring-up alongside the mission team.

Electronics Design Intern Sept 2025 – Jan 2026
SEDS Nepal (Remote) *Pokhara, Nepal*

- Engineered electronics modules for pseudo-satellite (HAPS) systems, ensuring modular and reproducible architecture.
- Performed component selection, BOM budgeting, and design reviews across interdisciplinary satellite subsystems.

Electronics Engineer Intern Jan 2025 – Jul 2025
Yarsa Tech *Pokhara, Nepal*

- Designed and optimized buck converter circuits; improved layout across multiple iterations using KiCad.
- Developed automated QC testing device for power backup products using C/C++ custom firmware.
- Built BLE-enabled QR/barcode scanner system; implemented complete firmware stack in C/C++.

Projects

LoRaNet Communication Module | *KiCad, ESP32, LoRa, C/C++*

- Engineered full schematic and multi-layer PCB for a long-range wireless system achieving ∼7 km LoS range.
- Authored firmware for reliable packet transmission over LoRa transceivers with ESP32 interfacing.

DC–DC Buck Converter | *KiCad, XL2013E1, Power Electronics*

- Architected a 12V-to-5V converter from schematic through PCB; iterated layout to achieve ∼90% efficiency and reduced ripple suitable for industrial use.

Micromouse Autonomous Maze-Solving Robot | *Custom PCB, PID, QTR Sensors*

- Fabricated a custom PCB reducing board size by 40%; achieved 95% navigation accuracy with PID motion control and a flood-fill maze-solving algorithm on an Arduino Nano.
- **1st Place** LOCUS 2025 (IOE Pulchowk), **2nd Place** Delta 5.0 (IOE Purwanchal), **5th Place** Techfest IIT Bombay.

Pocket CubeSat Module (1.5U PocketQube) | *Stacked PCB, LoRa, NodeMCU*

- Conceived stacked PCB architecture integrating OBC, power management, sensors, and LoRa comms in a 1.5U form factor with Li-Po solar charging.

Solar Greenhouse Automation System | *ESP32, BLE, MOSFET, OLED*

- Deployed solar-powered system monitoring soil, temperature, and humidity; automated irrigation via MOSFET control with BLE data transmission and real-time OLED dashboard.

Smart Helmet Safety System | *IoT, BLE, SpO2, Fall Detection*

- Prototyped a wearable device with pulse/SpO2 sensing, fall detection, and SOS alerts streamed via BLE to a supervisor monitoring dashboard.

Education

Pashchimanchal Campus (Tribhuvan University) Pokhara, Nepal
B.E. in Electronics, Communication & Information Engineering *2023 – Present (4th Year)*

Leadership

Electronics & Software Lead 2024 – 2026
Robotics Club, Pashchimanchal Campus

- Coached 350+ students in PCB design, firmware development, and robotics; launched an in-house PCB prototyping lab.
- Spearheaded competition robotics projects; organized workshops and technical training programs campus-wide, including a 15-day "Rainy Session" training 100+ students.

Achievements

Top 25 Nationwide – U.S.–Nepal Startup Weekend Challenge (Safety Modules), 2026
International Finalist – Techfest IIT Bombay, representing Nepal (Safety Modules), 2025
1st Place – LOCUS 2025 Micromouse, IOE Pulchowk Campus
Runner-Up – Delta Robothon, 2025
1st Runner-Up – Autonomous Boat Racing, Robotics Club WRC 2024
Winner – Battle For Speed / Line Follower Competition, Robotics Club 2023